Balanced Binary Tree (easy)

Problem Statement

Determine if a binary tree is height-balanced.

A binary tree is considered height-balanced if, for each node, the difference in height between its left and right subtrees is no more than one.

Examples:

1. **Input**:

```
    3
   / \
  9  20
     / \
    15  7
```

- **Expected Output**: true
- **Justification**: Every node in the tree has a left and right subtree depth difference of either 0 or 1.

2. **Input**:

```
        1
       / \
      2   2
     /     \
    3       3
   / \     / \
  4   4   4   4
```

- **Expected Output**: true
- **Justification**: Each node in the tree has a left and right subtree depth difference of either 0 or 1.

3. **Input**:

```
        1
       / \
      2   5
     /
    3
   /
  4
```

- **Expected Output**: false
- **Justification**: The root node has a left subtree depth of 3 and right subtree depth of 1. The difference (3 - 1 = 2) exceeds 1, hence the tree is not balanced.

Try it yourself

Try solving this question here:

